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STAYING CURRENT

Principles of interval training: a practical and physiological teaching guide for educators

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Abstract

We teach that physical activity, exercise, and exercise training are good for us and that the positive effect is better if we exercise with intensity; the greater the intensity, the greater the effect. Indeed, there is clear and irrefutable evidence of intensity dependence of effect, stemming from a dose-response relationship between effort going into the exercise and the result coming out of it. This leads to the situation where we generally and practically prescribe exercise as intensely as possible, for as long as possible, but this comes in conflict with the speed/power:duration relationship that dictates how long high-intensity exercise may be sustained. The solution we have is high-intensity interval training, referred to as HIIT or just interval training. This is a modality of exercise training that with repeated bouts of high-intensity exercise interspersed by active/passive recovery periods permits accumulation of intense exercise beyond that which would be achievable in a single exercise bout. This is then applied to a range of exercises and locomotion modes. However, while we often refer to and teach concepts of intensity dependence of exercise training effects, we rarely describe the basics of the actual exercise. Here, we introduce interval training, and by association, intensity, for the physiology-savvy, but nonexercise science educator. We conceptualize the principles of interval training and provide a practical framework and lesson plan for teaching appropriate interval training.

NEW & NOTEWORTHY We teach that we should do intense exercise. However, intense exercise is difficult to do and to teach. This poses a problem. Interval training is the practical solution. It offers a time-efficient way of carrying out intense exercise. However, teaching its principles requires nuance, which educators must be wary of when teaching sport and exercise science and physical education classes. This article provides a teaching- and educator-focused introduction to interval training.

exercise; intensity; intermittent; interval; training

TEACHING OBJECTIVES

In this article, we introduce interval training and, by association, intensity. This includes the principles and applications of interval training; design, implementation, and use thereof; and the underlying relevant exercise physiology. This is partly motivated by our observations of some common misconceptions among both students and educators: that interval training is prohibitively "hard" and unpleasant, that intensity needs to be maximized to an all-out effort (the "no pain-no gain" belief), that one format is "better" than others, and that only fit people can practice it. In addition, there are also conceptual misunderstandings about what interval training is, when and to whom it applies, and how to start and implement interval training in practice. We contend that the latter is often the result of inadequate knowledge and experience among educators who teach exercise training and physical activity. The format and depth of this article make it applicable for educators at senior undergraduate and graduate levels in Physiology, Human Biology, Sport and Exercise Science, and Kinesiology programs, as well as professional programs that include Medicine, Nursing, Physical Therapy, Allied Health, and Physical Education and Vocational Strength and Conditioning, Instructor, and Personal Trainer programs.

INTRODUCTION TO INTERVAL TRAINING

Interval training is a method of exercise that consists of repetitive high-intensity exercise bouts of variable intensity and duration that are alternated by periods of lower intensities. The aim of the high-intensity bouts is to exercise to near exhaustion, while the aim of the lower intensity periods is to allow for recovery. The alternation between intensities then makes it possible to reengage and continue with more high-intensity exercise within the same training session. In this manner, interval training may be carried out in a range of subjects, from high-performing athletes to patients with overt disease. For many, this is a new concept, but in reality, the concept is not new. Interval training has been around for a long time. Historically, humans have applied high-intensity locomotion for work such as hunting for 1,000s of years and industrial tasks for 100s of years,

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while in a recreational sports capacity, athletes have for well over 100 yr methodically and strategically used exercises and training regimens where bouts of specific high-intensity exercises are repeated a specific number of times to improve endurance performance. Once the value of interval training was realized and the application somewhat perfected, athletes would carry out interval training in solitude and in secret, preventing others from replicating their training. In the most well-known early cases of the Finnish and Czechoslovak runners Paavo Nurmi and Emil Zatopek, respectively, interval training was used in preparations that resulted in championship gold medals and World records. From this point onward, its impact was clear. Within a physical activity context, we have also started to apply interval training to improve health and function in the last 50 yr (1). Anecdotal observations have continued to indicate that exercise training and physical activity by this method may confer benefits that surpass those from other training methods or result in comparable benefits in a time-efficient manner compared to programs that rely on longer and less intense training sessions. In fact, in a theoretical argument, it has been suggested that to reach maximal biological potential for endurance performance, interval training is required, whereas in contrast only 90% performance potential may be achievable with exercise training that lacks interval training (1). Therefore, it was no surprise that science caught up, and sport and exercise research studies have thereafter also for more than 50 yr used a variety of interval training methods and interventions in both clinical human (2, 3) and experimental animal studies (4, 5), identifying many of the physiological determinants of, and adaptations to, high-intensity exercise as well as interval training.

TERMINOLOGY

There are two interlinked terms that require attention, "high-intensity" and "interval training." There is no universally agreed definition of either, and any use of the terms may be context specific (6).

High Intensity

First, let us define "high intensity." Intensity relates to the effort or work required to complete a specific task, or simply put: how hard your body works to handle a pace or rate of workload. It exists on a scale from "low"- to "high"-intensity domains. Various systems are used to measure intensity, where some have been developed for use in athletic populations and require specialist equipment and others are for the general public and rely on subjective feeling or something as simple as the ability to talk during exercise (1, 6–11). In this article, high intensity refers to any intensity above specific metabolic thresholds such as lactate threshold or critical speed in running or critical power or functional threshold power in cycling (8, 9, 12–14) (Table 1, *Metabolic thresholds*).

Interval Training

From intensity, it follows that interval training, or more correctly, high-intensity interval training (HIIT) is the practice of repeating short bouts of high-intensity exercise (thus exercise that is above the aforementioned metabolic thresholds), with interspersing periods of recovery (Fig. 1) (6, 15, 16). To standardize, avoid confusion, and align with terminology used by practitioners of interval training, we use specific terms in this article:

- "On" bouts: the high-intensity efforts. They are performed above the sustainable/nonsustainable metabolic threshold. On bouts may last from seconds to many minutes.
- "Off" bouts: the interspersing recovery periods. They consist of rest of light exercise to allow the subject to recover enough to perform another On bout and thereby accumulate more exercise at high intensity. Off bout duration may also vary, and range from seconds to a maximum of a few minutes.

The main purpose of the "On/Off" cycle is to help the subject do more high-intensity exercise than continuous exercise without recovery periods would have allowed (15, 16). Breaking the exercise into manageable pieces thereby allows the training stimulus to be optimized or maximized.

RATIONALE AND DESIGN OF INTERVAL TRAINING

With interval training, the repeated reengagement of high-intensity exercise allows the subject to accumulate a high training dose in the preferred high-intensity zone. This builds physical capacity and largely determines the outcome of the exercise training. If the interval training is manipulated appropriately, it can target intensities or work-rates higher than those required for a particular objective or event, in order to build overcapacity (15, 16). This allows athletes to carry out tactical maneuvers and sprint finishes to win races.

Interval training is not "one size fits all." Different demands require different approaches. In this regard, the distinguishing characteristic of interval training is intensity. Since the speed- or power–duration relationship means that volitional exhaustion becomes predictable (8, 9), it follows that the maximal duration for On bouts is set by intensity: higher intensity leads to shorter durations and lower intensity (but still within the realms of high intensity) permits longer durations (Fig. 2). In fact, calculating the speed- or power-duration relationship before interval training allows the subject to accurately set an individualized On bout work-rate according to their own physiological capacity, in order to optimize training (13). However, effective interval training does not necessitate On bouts being performed to complete volitional exhaustion. This is a common misconception that dedicated and motivated subjects often hold, which is why determining On bout intensity, duration, and repetition before commencing a session is critical.

The Off bouts that follow On bouts also deserve attention, including when teaching interval training. This is where recovery occurs that from a physiological point of view permits the next On bout to be carried out. As indicated earlier, On bouts are nonsteady-state and nonsustainable. Thus, after each On bout, the body must clear metabolites from the active muscle and refuel for continued work (8, 9, 13, 14) (Table 1, *Metabolites*). Off bouts allow this. It has been well-established that this is best achieved by a continuation of active steady-state low-intensity muscle work rather than

Table 1. *Summary of optional key physiology concepts associated with interval training*

Optional Key Physiology Concepts
Metabolic thresholds
Metabolic thresholds separate between different intensity zones by distinguishing sustainable steady-state exercise from nonsustainable non-steady-state exercise, with measurements taken in exercising muscle or in humans; measurements may be based on capillary blood lactate: • Lactate threshold: exercise intensity where blood lactate first rises from resting baseline level (lactate threshold 1) or starts to accumulate at a more rapid rate (lactate threshold 2); and • Maximal lactate steady-state: the maximum intensity where lactate clearance equals production. Or based on power or work performed: • Critical speed in running or critical power in cycling; exercise duration above metabolic threshold intensity is limited in a predictable manner and tolerable duration of the exercise is set by the specific work-rate being performed, where the speed- or power–duration relationship is fixed to a hyperbolic function that may be accurately calculated from a series of fixed work-rate time-to-exhaustion tests; • Functional threshold power in cycling: a power-based estimation of sustainable cycling duration; and • Critical speed/power and functional threshold power are closely related to lactate threshold 2.
Metabolites
Metabolites produced in muscle during high-intensity exercise include lactate, pyruvate, H^+, K^+, P_i, adenosine diphosphate and adenosine monophosphate, reactive oxygen species, and amino acid byproducts, all of which perturb intramuscular metabolic and contractile homeostasis, and together with depletion of muscle glycogen and phosphocreatine result in exercise cessation.
Slow component of oxygen uptake ($\dot{V}O_2$)
Slow component of oxygen uptake ($\dot{V}O_2$) refers to the observation that during high-intensity exercise above metabolic thresholds, $\dot{V}O_2$ does not stabilize at steady-state but drifts upward until $\dot{V}O_{2max}$ is reached, despite the exercise being performed at constant submaximal workload. This will slowly, but unintentionally, increase the physiological stress and experienced intensity of the exercise, lead to a loss of exercise and muscle efficiency, and may result in premature fatigue, even when this was not the intention of the "On" bout or exercise session. The rate of the slow component reflects the intensity of the exercise; higher intensity leads to faster drifts.
Bioenergetic systems
Bioenergetic systems that generate adenosine triphosphate (ATP) to fuel muscle work consist of the following: • Aerobic oxidative intracellular and mitochondrial pathways that include glycolysis, citric acid cycle, electron transport chain and oxidative phosphorylation for oxygen-dependent breakdown of glucose, fatty acids (via β-oxidation), and amino acids; and • Anaerobic intramuscular phosphagen, phosphocreatine, and glycolytic pathways that rapidly break down primarily glucose and glycogen in an oxygen-independent manner. Intensity dependence is explained by the fact that during very high-intensity exercise, oxygen supply is inadequate, so anaerobic bioenergetics dominate, while during lower intensity exercise (but still within high-intensity domains), adequate oxygen supply permits aerobic bioenergetics to dominate. Hence, anaerobic bioenergetics dominate during short-interval training, while aerobic bioenergetics dominate during long-interval training. The shift from anaerobic to aerobic bioenergetics within 1–2 minutes of exercise is explained by the fact that stored ATP, phosphocreatine, and glycogen deplete, and oxidative ATP generation therefore takes over. Exercise at or around metabolic thresholds creates a different intramuscular and intracellular bioenergetic challenge that depletes local substrate availability and produces lactate and metabolic by-products including H^+ that lead to acidic conditions that overcome buffer capacity, which challenges mitochondrial function, motor unit recruitment, and results in peripheral fatigue.
Proexercise adaptations
Proexercise adaptations to high-intensity interval training are as follows: • Cardiomyocyte excitation-contraction coupling and intracellular calcium handling that improve cardiac inotropy, lusitropy, and chronotropy, and cellular hypertrophy, resulting in enhanced pump function; • Vascular endothelial nitric oxide generation and smooth muscle reactive oxygen species handling that improve vasoreactivity and vasodilation, resulting in enhanced blood flow conduction; • Red blood cell mass and hemoglobin, resulting in enhanced oxygen-carrying capacity and delivery; • Capillaries and myoglobin, resulting in enhanced local perfusion and diffusion; and • Muscle mitochondria, mitochondrial and cytoplasmic metabolic, and substrate availability, resulting in enhanced ATP production.

passive rest or other interventions (17–20). After both short and intense and long and less-intense exercises that lead to exhaustion (21–23), active recovery is also intensity-dependent and most efficient when performed at or just below the sustainable/nonsustainable threshold for continued exercise. Engaging in moderately intense active recovery in the middle of an interval training session may not be subjectively preferable, but it works. This is because continued muscle work secures a relatively high blood flow and tissue perfusion, which facilitates metabolite clearance and substrate re-fueling. This leads to recovery and allows the subject to reach their training objectives. As such, intensity and duration of the recovery Off bout also contribute to shape interval training and are part of the exercise stress that subsequently shapes compensation, adaptation, and the outcome of interval training.

Therefore, how does an interval training session look? Planning and designing sessions include critical decisions about several aspects. These include the following:

- On bout: intensity, duration, frequency;
- Off bout: active/passive recovery intensity, duration, frequency; this will reflect frequency of On bouts as each On is followed by Off;
- Session: duration, frequency, locomotor mode, terrain;
- Recovery between sessions;
- Block(s) of sessions: duration and frequency;
- Alignment of block(s) of sessions with events and objectives; and
- Monitoring, adjustments, individualization, periodization, and specificity for ensuring progression, and variability to maintain motivation.

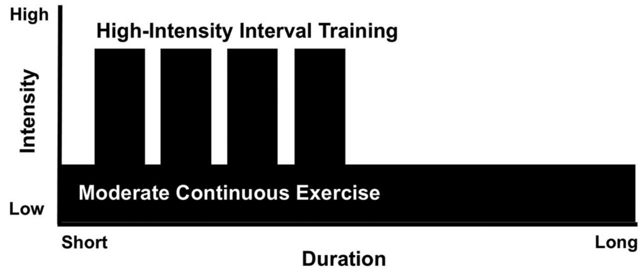

Figure 1. Schematic of an interval training session with interspersing high-intensity "On" bouts and recovery "Off" bouts, compared to moderate-intensity continuous exercise. How to teach with this figure: interval training and continuous exercise training are contrasted to highlight the fundamental differences with respect to intensity, duration, and session length. It should be emphasized that intensity, duration, and number of bouts may vary between different categories of interval training and between sessions of the same category, which is illustrated in Fig. 2.

Interval training may be applied to a range of locomotor modes, including walking, running, cycling, rowing, swimming, skiing, skating, and more complex forms of exercise or specific sports such as team-based sports, combat sports, climbing and mountaineering, aerobics, crossfit and fitness sports, and other sports. Examples of designs are shown in Figs. 1 and 2.

CATEGORIES

Interval training programs may vary due to individual factors such as fitness, recovery, nutrition, motivation, objectives, to which degree On bouts are performed to volitional exhaustion or not, available time and resources, environment, and personal preferences. However, interval training is in broad terms, typically classed into the below categories, each with their own characteristics and objectives. Conceptually, the main distinguishing factor is intensity and the consequence this intensity has on duration (Fig. 2), but in practical use, the description of each category of interval training tends to focus on duration or distance, or power if cycling, with the assumption that each bout is completed at or close to the intensity that the duration or power allows. Comparative characteristics of systematic interval training are listed in Table 2.

The set exercise intensity is tailored to individual maximal capacity (24), as especially highlighted in clinical practice involving the general public or patients with disease that are exercise training for health purposes, where the range of capacities or abilities may vary considerably (25, 26). This means that two subjects exercising next to each other may look very different from one another and may be exercising at very different absolute work rates, for example, one running fast and the other merely walking, yet they may be exercising at the same relative intensity, they may be experiencing the same relative effort, and they may be subject to the same physiological stress. The difference stems from different fitness levels. From this, it also follows that exercise capacity should preferably be measured in each individual before engaging with HIIT.

Long Interval

The most common, and the most advocated, interval training session consists of 4-min On bout durations repeated 4 times, colloquially termed 4-by-4 (or 4×4) interval or long-interval training. Typically, these long intervals have been performed as brisk uphill walking or running at intensities close to maximal heart rate (HR_{max}) or maximal oxygen uptake ($\dot{V}O_{2max}$), but not reaching either, for 4 min (range: 3–8 min) (13, 15, 24–28), whereas blood lactate may reach values >6 mM (29). Total On bout volume in a session is typically 10–30 min. The recovery Off bouts have included walking or jogging at considerably lower intensities, typically 2–4 min at 50%–60% of the On bout intensity. Cycling, cross-country skiing, skating, rowing, swimming, or other exercise modalities that involve dynamic work with the use of large muscle masses have also been applied using the same principles for intensity and duration (15, 30). Athlete populations preferring distance to duration, especially if exercising on tracks or courses, have translated this to

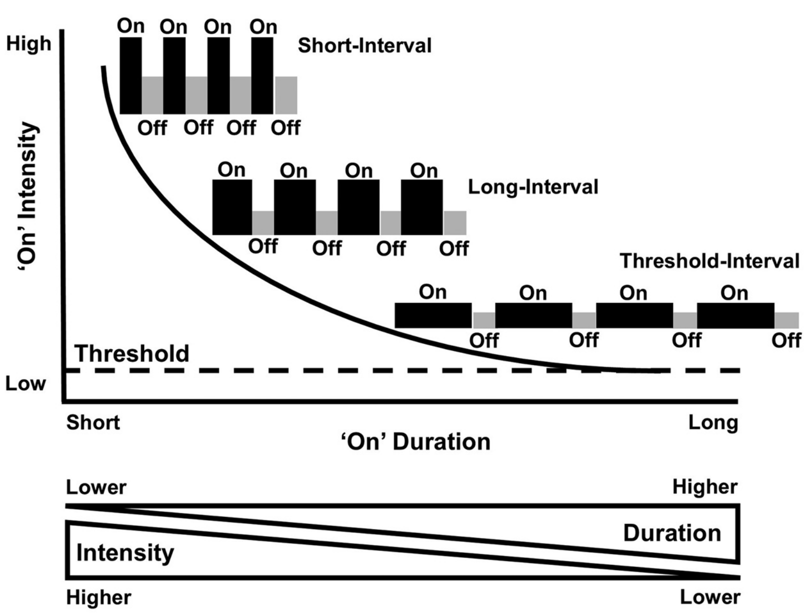

Figure 2. Schematic of interval training protocols where high-intensity "On" bout intensities on the left to right decrease from very-high/all-out sprint to exercise at or around metabolic threshold intensities that are lower but still in the high-intensity domain, while On bout sustainable durations correspondingly increase. How to teach with this figure: presentation follows from Fig. 1 that contrasted interval training to continuous moderate-intensity exercise training. Emphasis in this figure should be on the relationship between On bout intensities and sustainable durations; they are mutually dependent above the sustainable and nonsustainable exercise threshold (lactate threshold 2 or critical speed/power), and the relationship has a hyperbolic curvature. Examples include 4 On bouts per session, but this number may vary between different interval training programs.

Table 2. *Main characteristics of typical short-, long-, and threshold-interval training sessions*

	Short	Long	Threshold
Main objective	Anaerobic (and aerobic) capacity	Aerobic capacity, $\dot{V}O_{2max}$	Threshold and aerobic capacities
On intensity	All-out supramaximal	85%–95% $\dot{V}O_{2max}$ or HR_{max}	Around metabolic threshold
On duration	30–45 s	3–8 min	Variable, but with high total volume
On session volume	2–5 min	10–30 min	45 + minutes
Total session time	10–20 min	20–40 min	60 + minutes
Lactate	>6 mM with rapid accumulation	>6 mM	2–4 mM

How to teach with this table: the table contrasts the main design characteristics of short-, long-, and threshold-interval training sessions. A key emphasis is the main objective, which shapes the design, especially regarding intensity, duration, volume of "On" bouts, and total session time, while blood lactate may be measured to monitor success of the design, especially with respect to intensity. Advanced bioenergetic details of short-, long-, and threshold-interval sessions indicating the physiological and metabolic stress of the different categories of interval training are provided in *column 4*. Fartlek and sport-specific interval training protocols have not been included since they emulate short-, long-, and threshold-interval sessions in different environments. Stated values are approximations and context dependent. HR_{max}, maximal heart rate; $\dot{V}O_{2max}$, maximal oxygen uptake.

sessions consisting of, for example, 6 × 1,000-m bouts. If exercise testing for identifying maximal capacity and thereafter relative On bout speed or power for setting the preferred intensity before interval training is not possible, a practical rule of thumb has been that On bouts should cause "heavy breathing" but not "stiff legs." The classic physiological effect of long-interval training regimens is enhanced aerobic capacity, $\dot{V}O_{2max}$, and cardiovascular adaptations (15, 24, 27–31).

Threshold Interval

Threshold-interval training is a "close relative" of long-interval but with notable differences. The defining characteristic is that On bout intensity is at or around metabolic thresholds, typically lactate threshold 2 or critical speed (32). This allows On bouts to last longer or be repeated more frequently than the classical long-interval described above. Typical sessions may therefore consist of 4 × 3,000-m On bouts at half-marathon pace, or alternatively 25 × 400 meter at 600- to 800-meter pace, with 30–60 s Off bouts. If intensity is carefully monitored such that the training exerts a relatively low metabolic stress on the locomotor muscles, it avoids long postexercise recovery periods. This allows high-level athletes with extensive training backgrounds to undergo threshold-interval training sessions on consecutive days or even twice a day (double threshold days) on selected days if recovery days are inserted in between. The result is that threshold-interval training permits a high training volume (total amount of exercise), with On bout volume in individual sessions typically lasting 45–60 min or even longer and has led to considerable adoption and success in elite-level middle-, long-, and ultra-long-distance endurance athletes (33). However, for appropriate intensity monitoring, On bout pacing is generally blood lactate-guided, with acceptable concentrations typically ranging 2–4 mM (32), which is substantially lower than the target value for long-interval training cited above. Since much of the basics of threshold-interval training was described by Norwegian athletes and practitioners and since the method applies intensity distributions that have developed beyond previously established models (34, 35), this method of interval training has latterly been popularized as the "Norwegian Method" of endurance training (36). The classic physiological effect of threshold-interval training is also enhanced aerobic capacity, $\dot{V}O_{2max}$, and cardiovascular adaptations, similar to, but with a lesser magnitude of effects, than long-interval training, but also considerable skeletal muscle adaptations and notably, raised metabolic threshold and increased threshold speed and power, due to improved skeletal muscle metabolite and lactate buffering, substrate availability and utilization, and fatigue resistance (15, 32, 33, 37, 38).

Short Interval

Compared to the aerobic-focused long- and threshold-interval training methods, short-interval training is in some way the diametrical opposite. However, it too has gained practical traction and scientific attention and has been found to result in considerable performance and physiological adaptations. Short-interval training consists of exercise On bouts characterized by anaerobic sprint-type maximal (or supramaximal) efforts of very short durations, typically 30 s, with a range of 10–45 s. Short-interval training is therefore also referred to as sprint-interval training (SIT). These bouts are repeated consecutively as ∼5–10 or more On bouts, separated by recovery Off bouts lasting seconds to minutes of passive or low-intensity exercise active recovery (14, 16, 28, 39, 40). These sets may also be repeated two to three times, depending on the number of On bouts and duration of Off bouts, with prolonged recovery periods between sets. An often applied approach is the use of repeated anaerobic 30-s all-out Wingate cycling tests, with a typical training session consisting of four to six repeats separated by 2–4 or more minutes of recovery. This accumulates a total On bout time of 2–5 min consisting of very intense work per session and a total session time of less than 20 min, which makes it very time-efficient. As a result, short-interval training may have a highly reduced training volume, often only 10%–20% to that of other training. Blood lactate concentration may reach >6 mM, which may not be higher than the equivalent in long-interval training, but during short-interval training, peak lactate is reached in considerably shorter time, and therefore the rate of lactate accumulation is higher. Although these On bouts are short, the sessions are demanding, may not always be tolerable or even safe for subjects at risk, or may not be appealing to everyone, but in some cases, it has been found that short-interval training may compare well to more traditional high-volume endurance exercise training regimens (41). In contrast to long- and threshold-interval training, short-interval training cannot be said to result in "classic" physiological effects, because the method is rather new and there is nothing "classic" (as in historical) about it, but the described effects include enhanced aerobic capacity and $\dot{V}O_{2max}$ (34, 39, 42, 43), although the magnitude of effect tends to be smaller than the effect of long- or threshold-interval training. However, scaled to exercise training volume, the effect of short-interval training may be substantial. In contrast to the cardiovascular and central effects of interval training with longer On durations (15, 24, 27–31), short-interval training mainly improves peripheral skeletal muscle oxidative capacity, favors substrate shifts that enhance muscle tissue energetics, and increases capillarization, in addition to increasing anaerobic metabolic capacity (14, 16, 34, 39, 42).

Fartlek

Fartlek comes from Swedish and means "speedplay" and was introduced to exercise training in practice at least ~100 yr ago, originally as a form of continuous running with alternating but unsystematic periods of faster and slower paces, usually with no or few stops and carried out in nature (1). In nature, the natural ground and surface variation lends itself to fluctuation in speed, effort, and intensity. Today, fartlek also applies to other locomotor modes such as cycling and cross-country skiing. In contrast to other forms of interval training, focus tends to be on speed or power rather than more formal metrics of intensity and is often utilized by beginner or recreationally active subjects who want some speed or power work or higher intensities incorporated into their training sessions, but who also prefer flexibility and control to the rigor of other forms of interval training. To many, fartlek also adds joy. Some may however question if fartlek is interval training at all, and if it is, is it any good? To this, and partly accentuated by fartlek not being standardized, there is no simple answer, apart from stating that it can be but may not be. To some, it will be too unstructured and lacking the clear objectives that other forms of interval training offer, while for others, it is an opportunity to harvest some of the benefits of interval training without fully embracing it. Notwithstanding this, appropriate fartleks may increase the physiological stress to the body in a manner comparable to more systematic interval training, albeit with less magnitude, especially if applied with more systematic insertions of specific speed or power periods (44, 45). Alternatively, the fartlek has also been introduced to control groups as a contrast to other and more specific forms of interval training (46). Because intensity is normally not quantified or standardized during fartlek training, it is difficult to denote any specific adaptations to it, but they probably reflect the nature of the exercise training that has been completed. For example, some speed or power work will likely result in some intensity-dependent effects that originate from the chosen, but unbeknownst, intensities, but because appropriate intensity monitoring and control are usually lacking, the magnitude of effect may be limited.

Sport-Specific Interval

Interval training may also be applied in more team-sport-specific settings, for example, during modified game play in football, hockey, basketball, and other sports. Key factors for implementing this would first be clear communication of expectations to players, where the objective is gameplay at high-intensity, and modification to the game by adjusting (usually limiting) duration, pitch size, and number of active players, frequent use of substitutes, altering game rules, make use of walls, side-flipped benches or surplus players acting as "side walls" to keep ball in play, and immediately replace game balls that go out of play with spare balls (47). Many of these implementations are often achieved by shorter duration small-sided games as part of longer team training sessions (48–50), where the aim is to mimic long-interval training; however, the exact design of the session also depends on the number of available players. Typically, On and Off bouts are equally long if players split into four teams alternate between play and recovery, or On bouts are twice as long as Off bouts if players are split into three teams with two On and one Off bouts in a full cycle. While the above examples come from soccer, other sports have also utilized sports-specific small-sided games that incorporate interval training principles, including basketball (51), rugby (52), and wheelchair sports (53). A different, but still sport-specific, approach has also been the adoption of on-pitch ball dribbling courses where different route segments, obstacles, and tasks challenge the player in a sport-specific manner. Examples include dribbling with close ball control around cones, where individual players take turns at a pace that mimics long-interval training sessions (48, 54). Both small-sided games and sport-specific courses have been effective for increasing endurance capacity and $\dot{V}O_{2max}$, with comparable effect sizes to more traditional interventions where similar players have carried out regular running long-interval training drills as part of team training sessions (55). Short-interval training may also be emulated by small-sided games or courses in team settings, with the added benefit that this may also reflect the intermittent activity pattern typically observed in team sports, with interspersing very high-intensity sprint and very low-intensity standing or walking periods (1).

MONITORING

Because relative intensity is key, appropriate testing of parameters that can be used for intensity prescription often comes first (28, 30) and is therefore also included in the intended learning objectives (ILOs) for students. This may be an incremental exercise test measuring $\dot{V}O_{2max}$, HR_{max}, maximal aerobic speed or power, lactate threshold, or other performance metrics like critical speed or power, functional threshold power, or blip/performance-tests in a laboratory or on the field, using the same locomotor mode as the subsequent training (56). The results may be used to set training intensity. A well-controlled program will then retest during and upon the conclusion of the interval training program to monitor progress and adjust absolute workload to maintain the set relative intensity when exercise capacity improves. After the appropriate workload for a given intensity has been established, this should then be monitored during the course of training to ensure that targets are realized. This may be via real-time measurements of heart rate (HR), blood lactate, or performed workload or subjective measures like rate of perceived exertion (RPE) or feeling of effort and strenuousness (15, 16, 30, 32). For beginner and recreational subjects, it will be important not to start a session with, for example, long- or threshold-interval On bouts that are too intense, since the slow component of oxygen uptake ($\dot{V}O_2$) kinetics will increase the physiological stress during the course of an On bout, despite a constant workload (13, 57) [Table 1, *Slow component of oxygen uptake ($\dot{V}O_2$)*]. For most subjects, On bouts should typically end when the subject still has residual capacity and feels one to two On bouts might yet have been achievable. Overextending the exercise into complete volitional exhaustion is contraindicated since this may lead to excessive fatigue that requires prolonged recovery and increases injury risk (57). The effect of active versus passive recovery during Off bouts of interval training could be examined by completing two separate sessions with similar On bouts; in one session, the recovery Off bouts would be passive (complete rest), and in the other, the recovery Off bouts would involve moderate-intensity exercise, where exercise responses and the ability to perform and complete the sessions would be compared.

INTERVAL TRAINING IN PRACTICE

Interval training is a learned practice. This means that when teaching interval training, practical guidance about how to carry out sessions should also be offered. Beginner subjects are advised to optimize the interval training protocol rather than "maximize" it and learn by trial-and-error in order to establish practices that are manageable in the long term. They should also consider supervised, rather than unsupervised training, as this tends to increase adherence to any training that involves high-intensity (58). In high-level athletes, however, pushing the intensity toward the ceiling of the intensity zone may in contrast be beneficial to increase the exercise adaptation (59). Regardless of fitness level and previous history of exercise training, it is important to make interval training part of a more holistic approach with regard to intensity distribution. If little overall exercise training is performed, such as one to two sessions per week, both may potentially involve interval training, but if the total volume exceeds this, some sessions should focus on exercise with high intensity and others on low-intensity. Several models for optimizing intensity distributions have been developed, but common to all is that when exercise training volume is high, only a minority of the training should be at high intensity (34, 35, 60). Even elite endurance athletes train with high intensity for only a small proportion of their total exercise training volume (61) and typically in periodized blocks that are planned into performance preparation macrocycles (29, 32, 33). However, despite interval training being unpleasant to many because of the high effort, subjects of various backgrounds and with different training histories report that it is acceptable and feasible (58) and more rewarding than other forms of training because of the superior effect (26).

We note the following lived experiences on intensity when teaching interval training:

- During short-interval training, intensity will drop during the session; this is expected;
- During long- and threshold-interval training, appropriate intensity monitoring is key;
- Long-interval training, and especially for beginners, do not start too hard; rather, aim to reach intensity zone late in On bout or within two to three On bouts, otherwise intensity will drop prematurely;
- Threshold-interval training, it is often preferable to titrate On bout intensity to just below threshold rather than at or above threshold;
- Fartlek is often a desirable choice for those using moderately hilly areas such as parks or when the course is set by, for example, active commute;
- Sport-specific interval training, intensity is often individual player-dependent, where some respond positively to intensity objectives, and others will need reminding of desired intensity because they mentally tune to different game aspects such as technical/tactical; and
- Interval training is subjectively hard; if your subject is feeling ill, not rested, fatigued, or otherwise ill prepared, the bar for changing or even cancelling/postponing a session should be low.

BIOENERGETIC EFFECTS OF INTERVAL TRAINING

The energy for muscle work during interval training comes from aerobic and anaerobic sources that provide the necessary adenosine triphosphate (ATP), with their specific contributions dependent upon intensity; lower intensity is largely aerobic, while higher intensity has a larger anaerobic contribution (Table 1, *Bioenergetic systems*). There is also time dependence, as during intense exercise, a shift from mainly anaerobic to mainly aerobic energy contributions occurs within the first 1–2 min of exercise (62, 63). When this happens, exercise intensity inevitably drops. Because of the close relationship between interval training and bioenergetics, it becomes important that this link between different protocol designs and the bioenergetics including fuel availability, is adequately included when teaching and learning interval training; otherwise, it would be difficult to conceptually and fully appreciate why interval training works and why the designs of different protocols are the way they are. Therefore, bioenergetics concepts will also be included as part of the ILOs.

Different intensities during On bouts stress different aspects of aerobic and anaerobic bioenergetics. To improve aerobic capacity and $\dot{V}O_{2max}$, exercise training needs to tax physiological systems that contribute to oxygen transport and utilization, especially cardiovascular conductance and skeletal muscle perfusion and metabolic capacities (6, 15, 27, 64). This is achieved by long-duration high-intensity interval training, due to the slow increase kinetics of $\dot{V}O_2$ and cardiac output at the start of each On bout that necessitates long durations. In addition, this will also result in the accumulation of excess postexercise oxygen consumption (EPOC) that further adds to the physiological stress (65). In contrast, different principles apply for anaerobic bioenergetics during very high and supramaximal exercise. Since these sources are limited, this effort may only be sustained for short periods of time, such as 10–45 s (14, 16). Adaptation in these systems is therefore provoked by short-interval training (39, 40). Thus, there is solid justification to apply both short and long intervals as part of regular exercise training, because they result in different physiological and metabolic adaptations. Fartlek and sport-specific interval training may also include all of this since they may contain the same intensities (1). However, what about threshold-interval training? These On bouts are executed at relatively low metabolic cost, so the majority of energy comes from aerobic processes fueled by adequate oxygen supply, but this challenge is less than that of long-interval training, and therefore cardiovascular adaptations will be of lesser magnitude and take longer to manifest. Instead, sustained work around metabolic thresholds is often characterized by development of peripheral fatigue (32, 37, 38), and the prevailing adaptation is increased exercise tolerance and fatigue resistance in working skeletal muscle at high, but submaximal intensity exercise.

ORGAN SYSTEM EFFECTS OF INTERVAL TRAINING

Interval training improves function along the oxygen transport cascade in a manner that improves exercise capacity, tolerance, and fatigue resistance. The stressor that results in physiological adaptation is acute exercise (66), and since interval training exerts a high physiological stress, it follows that the magnitude of the training effect is concomitantly larger with interval training than exercise training with lower intensities (13, 24). The evidence so far indicates that long-interval training is a major contributor to central cardiovascular adaptations (24, 27, 30, 31, 67–70), due to the high metabolic demand exerted by exercise in the working skeletal muscles and the consequential demand to oxygen supply, while short-interval training is a major contributor to peripheral skeletal muscle adaptations, due to the inadequate supply that consequentially develops anaerobic conditions in the working skeletal muscles (14, 39, 40, 42). The result of this is cellular and tissue adaptations in key organ systems that facilitate uptake, conductance, perfusion, and utilization of oxygen under aerobic conditions and local working muscle ATP generation under aerobic and anaerobic conditions (Table 1, *Proexercise adaptations*), which therefore also are included in ILOs.

IN HEALTH AND DISEASE

The principles described apply in both health and disease, and the intended readership of this article also includes those training for or are already in clinical practice with patient clients. We therefore point out that physical activity that also involves high-intensity exercise is recommended for prevention, treatment, and rehabilitation in various clinical populations with heightened risk of developing dysfunction and disease or where overt dysfunction or disease has already presented itself (68–70). High-intensity exercise in this scenario is best achieved by interval training, similar to that of healthy individuals. As such, interval training has been introduced to models and patient groups with pulmonary, cardiovascular, skeletal muscle, and neurophysiological, metabolic, endocrine, cancer, and mental disorders and diseases, and importantly, interval training benefits have been shown to extend to also those scenarios (1, 6, 24–26, 30, 68–72). In these studies, accommodations to individual health, function, and circumstances of patients have been made, patient safety has been monitored and maintained, and the principles of interval training have been followed. The results have broadly shown that higher intensities of exercise such as those achieved during interval training have conferred greater impacts than training with lower exercise intensities or have conferred similar impacts to training with lower exercise intensities but with use of less training time and volume. These impacts have included reversal of dysfunction as well as improvements to functional work capacity, mental health, and quality of life. An argument may also be made that long-term interval training may prolong life, because it increases $\dot{V}O_{2max}$ and $\dot{V}O_{2max}$ is inversely linked to mortality (73), although the relationship between interval training and mortality has not been directly assessed. Nonetheless, this means interval training may be used in prevention, therapy, and rehabilitation.

IMPLEMENTATION TO TEACHING

Interval training can be taught in the classroom, laboratory, clinic, indoor sports halls, and in outdoor fields, as part of teaching and instruction in Sport and Exercise Science, Kinesiology, Human Physiology and Biology, Physical Education, Medicine, Nursing, Physical Therapy, and Allied Health courses. The below lesson plan is designed for senior undergraduate and graduate levels but may be adapted for applied and vocational Strength and Conditioning, Instructor, or Personal Trainer courses. We suggest teaching interval training and high-intensity exercise as a class activity that integrates the theoretical basis delivered in the classroom with practical instruction in the laboratory or sports field, the latter of which may be adapted to various resource availabilities.

Lesson Plan

Classroom activity.

The classroom activity includes the following:

- ILOs: By the end of session, students should be able to describe interval training and high-intensity exercise, differentiate between different categories of interval training, design practical interval training sessions that include high-intensity On and recovery Off bouts and that are designed to reach specific training objectives, monitor and evaluate intensity during interval training, describe physiological responses and chronic adaptations to interval training in associated physiological and bioenergetics systems, and describe how different interval training designs stress and adapt physiological and bioenergetics systems differently;
- For all levels, the above ILOs are purposefully designed to cover both "what" (what is interval training and how may it be practiced) and "why" (why do we practice

interval training and what are the effects) of interval training;
- For advanced levels of study, ILOs may extend to include molecular mechanisms of physiological adaptations;
- For advanced and applied or vocational programs, ILOs may extend to include designs of micro- to macrocycle programs of exercise training that incorporate interval training;
- For clinical programs, ILOs may extend to include application of interval training in patient populations; and
- Format: lecture/seminar; duration: 120 minutes for teaching basic principles, and 60–120 minutes for teaching advanced, applied, or clinical principles.

Preparation for practical activity, delivered either in conjunction with classroom activity or as a separate information package.
The preparation for the practical activity includes the following:

- Discuss and decide interval training exercise protocol including exercise modality, with emphasis on intensity, duration, and appropriate monitoring;
- Establish session objectives; and
- Student participants required 1–2 days before activity: avoid alcohol and strenuous exercise, and 2–3 hours prior to activity: eat normal meal, hydrate, and prioritize rest.

Practical activity.
The practical activity includes the following:

- Continue hydration, consume electrolytes and energy as appropriate; wear comfortable clothing; if possible, avoid heat and sun exposure, high humidity, and inclement weather conditions;
- Complete participation consent; for safety clearance, complete health screens, and readiness forms (74);
- Instructor to demonstrate monitoring techniques and manage expectations: interval training may feel uncomfortable, and participants should expect breathlessness;
- Recapitulate exercise protocol and session objectives;
- Warm-up: stretching/activation exercises using body weight or elastic bands; low-intensity exercise in the chosen modality;
- Interval training: follow exercise protocol including high-intensity On and recovery Off bouts, focus on achieving correct intensity and duration;
- Monitoring: measure exercise stress as appropriate, which may include feeling of effort level, RPE or other perception metrics, HR, $\dot{V}O_2$, blood lactate, sweat, body temperature, and performance metrics commensurate to exercise modality;
- Postexercise: rehydrate and replenish with carbohydrates and proteins; active or passive recovery and rest; consider stretching, cold-water immersion/ice baths, or massage; and
- If several sessions are available: schedule and complete different categories of interval training, and complete separate interval training sessions with similar On bouts but different Off bouts, for example, active versus passive recovery, to demonstrate and experience the breadth of interval training protocols.

Postpractical classroom activity.
The postpractual classroom activity includes the following:

- Debrief: how did session feel, how did it compare against expectations, how difficult it was to adhere to the prescribed exercise protocol, discuss recovery; and
- Analyze and interpret results from monitoring; formal and informal evaluation of session; assess whether session objectives were achieved; establish practical learning points.

Sample discussion and learning points would center around effort and intensity, expectations and realizations following interval training, adherence to and compliance with exercise protocol, feasibility and safety of interval training for specific cohorts, and achievement of session objectives. Special consideration should be given to issues arising from starting On bouts too intensely, ensuring Off bouts are active rather than passive, feasibility of supervised versus unsupervised exercise, and factors that may motivate short- and long-term participation, adherence, and compliance. Adherence and compliance is particularly important to establish and discuss in clinical classes and where interval training is used for clinical purposes, since interval training has shown conceptual promise in clinical cohorts with overt disease where exercise training has been well-controlled and supervised (24–26, 30), but in large-scale home-based implementations without supervised exercise training in the same patient cohorts, interval training has mainly failed, at least partly due to low adherence and compliance with protocols (75, 76). For assessment purposes, it would be appropriate to focus on design, justification of design, implementation, and monitoring of interval training, together with exercise physiology-centered discussion of responses and adaptations to interval training, either via submission of a report or as part of an exam.

CONCLUSIONS

Interval training has become a popular and time-efficient method for accumulating high-intensity exercise to increase physiologic adaptations. As such, the principles and practical application of interval training have also found their way into undergraduate and graduate, and professional and vocational, curricula. Preparation of instructors has, however, lagged behind. This article has equipped the instructor with the theoretical and practical background for introducing and implementing interval training into teaching and practice, including the provision of a lesson plan with classroom and laboratory or field activities that support this teaching, because impactful teaching of interval training requires focus on both theoretical and practical aspects. Therefore, we have also developed practical guidance to meet the demands of those new to interval training.

KEY TAKEAWAYS

The key takeaways from this study are as follows:

- Interval training has long been established as an exercise training method in endurance sports, with many successes throughout the history of sport and physical activity;
- Interval training uses repeated bouts of high-intensity exercise interspersed with rest or light exercise recovery periods (On and Off bouts);
- High-intensity interval training allows a greater total duration of work performed at high intensity than continuous exercise as recovery periods delay fatigue and allow repeated efforts;
- Different protocols of interval training use specific intensities that shape the protocol and determine acute physiological responses and chronic adaptations;
- Interval training is key for optimizing performance in endurance sports but is also used to improve health and functionality in noncompetitive cohorts including patients; and
- This article has presented a framework for teaching practical and theoretical principles of interval training to undergraduate, graduate, professional, and vocational study programs.

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DISCLOSURES

No conflicts of interest, financial or otherwise, are declared by the authors.

AUTHOR CONTRIBUTIONS

O.J.K. conceived and designed research; R.I., V.R., and O.J.K. performed experiments; R.I., V.R., and O.J.K. analyzed data; R.I., V.R., and O.J.K. interpreted results of experiments; R.I., V.R., and O.J.K. prepared figures; R.I., V.R., and O.J.K. drafted manuscript; R.I., V.R., and O.J.K. edited and revised manuscript; R.I., V.R., and O.J.K. approved final version of manuscript.

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